

Sergio Insuasti

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Summary

Computer Science student graduating in 2026 from The University of New South Wales, with a portfolio of full-stack and systems-level projects in languages such as Python, Java and SQL. Thrives in collaborative environments and seeks a software engineering role to contribute clean, reliable code and make a real-world impact through technology.

Education

University of New South Wales

Feb 2024 – Dec 2026

Bachelor of Computer Science (Distinction)

- Relevant Coursework: Data Structures & Algorithms, Software Design & Architecture, Software Construction, Algorithm Design and Analysis, Programming for Data Analysis
- Notable Highlights: Software Design & Architecture: 90 HD, SWE Fundamentals: 87 HD, Programming for Data Analysis: 85 HD

Technical Skills

Programming Languages: Python, Java, SQL, C, C++, Javascript, Typescript, R, Stata, HTML, CSS, Bash

Frameworks & Tools: Git, React, Spring Framework, Docker, AWS, Google Cloud, Apache Kafka, Rest API

Experience

UNSW Artificial Intelligence Society

Projects Subcommittee

Feb 2026 -

Website Update and Maintenance

- Implemented an Admin Dashboard using Firebase Auth for event submission/creation
- Incorporated NoSQL and Bucket Storage (Firestore and Firebase Storage) for event persistence
- Automated Team Notification System using Discord Webhook API
- Automated event uploading through Firebase's SDK reducing 10+ event submission times by 5 hours

UNSW Wayfinder

- Rendered UNSW Campus in an interactable 3D environment
- Implemented 3D Gaussian Splatting to render 2D images into 3D scenes
- Customised path selection for users (e.g. Least Steps, Highest Climb)

Personal Projects

Patient Management System

- Implemented a production-ready backend system designed with Microservices architecture
- Modelled persistent relational databases for user authentication and patient records using PostgreSQL
- Included industry-level software implementation such as Docker and AWS
- Guaranteed data integrity, role-based access control through Spring Security, PostgreSQL and JWT
- Enforced clear separation between synchronous API calls and asynchronous event-driven workflows
- Implemented system security through JWT token authentication

Options Pricing Educational Dashboard

- Implemented multiple options pricing models with cross-model comparison and caching for faster computational performance
- Developed a responsive system that recomputes models simultaneously to user changes
- Executed high-level decisions regarding complexity, frontend integration and system design, including refactoring for separation of concerns and software design principles

Extra-Curricular Activities

University Societies at UNSW: Artificial Intelligence Society (Internals), Computer Science and Engineering Society, Data Society, Economics Society, Boulderling Society, Matcha Society, FinTech Society, Competitive Programming and Mathematics Society

Additional Activities: Music Producer, Composer and Audio Engineer

References:

Available on request